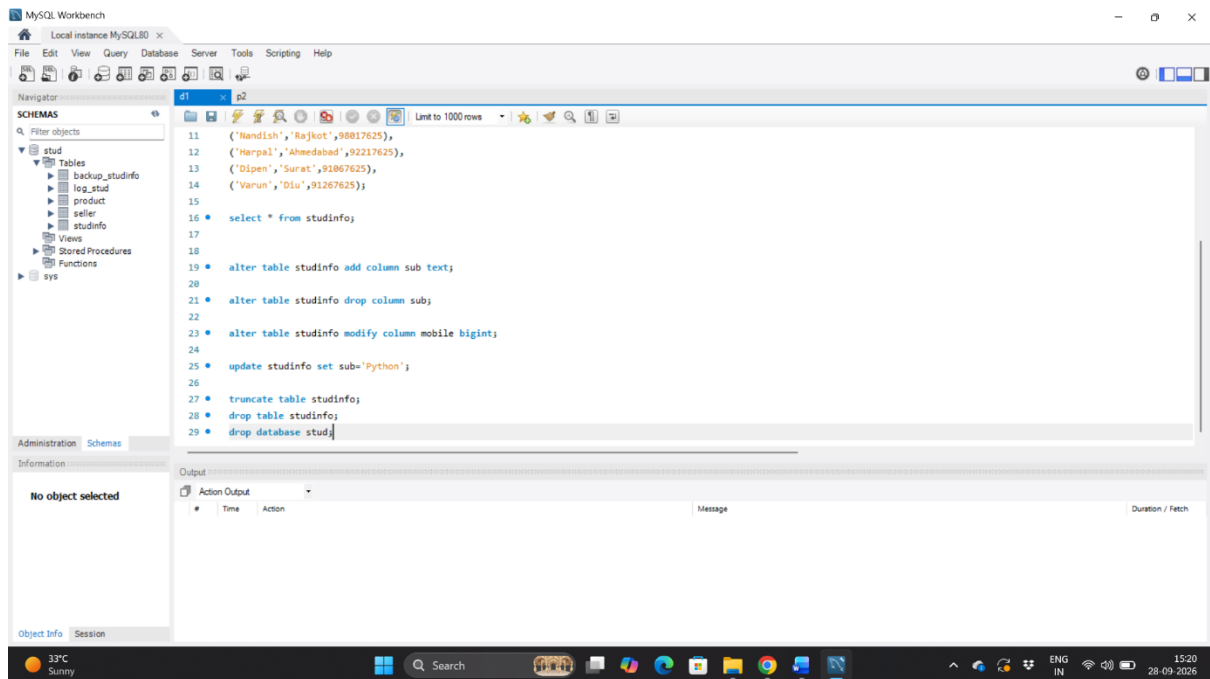


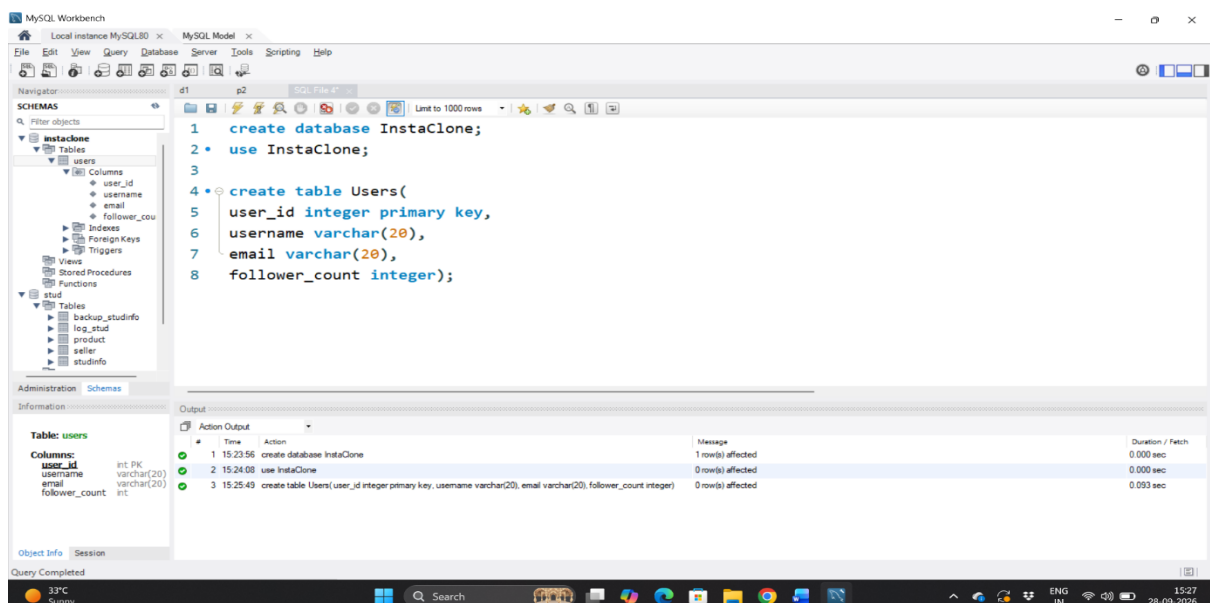
Session 1

Q1. Install MySQL Workbench on your computer, connect to the local MySQL server, and take a screenshot of the default databases shown in the Schemas panel.

Ans:



Q2. Create a new database called InstaClone in MySQL Workbench, then create a table named Users with columns: user_id (INT, primary key), username (VARCHAR), email (VARCHAR), and followers_count (INT).



Q3. Insert 3 sample users into the Users table you created, using realistic Instagram-style usernames and follower counts.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following code:

```

10 insert into Users(username,email,follower_count)values
11 ('purvang_kb059','purvangkb@gmail.com',335),
12 ('viratkohli','virat05@gmail.com',1200),
13 ('iasrk','shahrukhkhan@gmail.com',1300);
14
15 select * from Users;
16

```

The Results grid shows the output of the query:

user_id	username	email	follower_count
1	purvang_kb059	purvangkb@gmail.com	335
2	viratkohli	virat05@gmail.com	1200
3	iasrk	shahrukhkhan@gmail.com	1300

The Action Output pane shows the execution details:

#	Time	Action	Message	Duration / Fetch
6	15:33:37	Drop table Users	0 row(s) affected	0.031 sec
7	15:33:44	create table Users(user_id integer primary key auto_increment, username varchar(20), email varchar(20), follow...	0 row(s) affected	0.078 sec
8	15:33:48	insert into Users(username,email,follower_count)values (purvang_kb059,'purvangkb@gmail.com',335), (viratk...	Error Code: 1406. Data too long for column 'email' at row 3	0.063 sec
9	15:34:17	Drop table Users	0 row(s) affected	0.016 sec
10	15:34:26	create table Users(user_id integer primary key auto_increment, username varchar(220), email varchar(220), foll...	0 row(s) affected	0.032 sec
11	15:34:36	insert into Users(username,email,follower_count)values (purvang_kb059,'purvangkb@gmail.com',335), (viratk...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
12	15:34:52	select * from Users LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec

Q4. Create another table in the InstaClone database called Posts with columns: post_id (INT, primary key), user_id (INT), caption (VARCHAR), and post_date (DATE). Add a foreign key from Posts.user_id to Users.user_id to establish a relationship.

The screenshot shows the MySQL Workbench interface. The SQL editor contains the following code:

```

13 ('iasrk','shahrukhkhan@gmail.com',1300);
14
15 select * from Users;
16
17 CREATE TABLE Posts (
18     post_id INT PRIMARY KEY auto_increment,
19     user_id INT,
20     caption VARCHAR(255),
21     post_date DATE,
22     FOREIGN KEY (user_id) REFERENCES Users(user_id)
23 );

```

The Action Output pane shows the execution details:

#	Time	Action	Message	Duration / Fetch
7	15:33:44	create table Users(user_id integer primary key auto_increment, username varchar(20), email varchar(20), follow...	0 row(s) affected	0.078 sec
8	15:33:48	insert into Users(username,email,follower_count)values (purvang_kb059,'purvangkb@gmail.com',335), (viratk...	Error Code: 1406. Data too long for column 'email' at row 3	0.063 sec
9	15:34:17	Drop table Users	0 row(s) affected	0.016 sec
10	15:34:26	create table Users(user_id integer primary key auto_increment, username varchar(220), email varchar(220), foll...	0 row(s) affected	0.032 sec
11	15:34:36	insert into Users(username,email,follower_count)values (purvang_kb059,'purvangkb@gmail.com',335), (viratk...	3 row(s) affected Records: 3 Duplicates: 0 Warnings: 0	0.000 sec
12	15:34:52	select * from Users LIMIT 0, 1000	3 row(s) returned	0.000 sec / 0.000 sec
13	15:39:42	CREATE TABLE Posts (post_id INT PRIMARY KEY auto_increment, user_id INT, caption VARCHAR(255), post_date DATE, FOREIGN KEY (user_id) REFERENCES Users(user_id));	0 row(s) affected	0.047 sec

Q5. Write a short comparison (3-4 lines each) of MySQL, PostgreSQL, Oracle, and SQLite, focusing on where each one is commonly used (for example: web apps, mobile apps, enterprise systems, etc.).

Ans:

MySQL is widely used for **web applications** and websites.
It is commonly used with PHP, WordPress, and e-commerce applications.
It is suitable for applications that need a reliable relational database.

PostgreSQL

PostgreSQL is an advanced open-source relational database.
It is commonly used for **web applications, analytics, and data-heavy systems**.
It is known for strong SQL support and advanced data types.

Oracle

Oracle Database is widely used in **large enterprise systems**.
Banks, governments, and large organizations use it for business-critical applications.
It is designed for large amounts of data and complex database operations.

SQLite

SQLite is a lightweight database that runs directly inside an application.
It is commonly used in **mobile apps, desktop applications, and embedded systems**.
It does not require a separate database server